

Juan Jude Pereira

Software Engineer | AI/ML | Generative AI

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GitHub | LinkedIn | Website

SUMMARY

Computer Science graduate with hands-on experience in AI engineering, NLP, Retrieval-Augmented Generation (RAG), Agentic AI, and full-stack development. Skilled in Python, Machine Learning, Generative AI, React, JavaScript, Node.js, SQL, Pandas, NumPy, and Streamlit, with experience developing practical AI and software solutions through internship and project-based work.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, SQL

Full-Stack Development: HTML, CSS, React, Node.js

Databases: MySQL, MongoDB

AI/ML: Machine Learning, NLP, Generative AI, Retrieval-Augmented Generation (RAG), Agentic AI, Large Language Models (LLMs), Qwen

Data & Tools: Pandas, NumPy, FAISS, Streamlit, Git, GitHub

EXPERIENCE

AkumenbyQ

Artificial Intelligence Intern

Jul 2026 – Present

- Engineered a Hybrid Retrieval-Augmented Generation (RAG) pipeline combining BM25 keyword retrieval and semantic vector search to enhance document retrieval and response relevance.
- Implemented Agentic AI workflows for task decomposition, information retrieval, context management, and AI-driven response generation.
- Built an interactive Student Performance Dashboard using Python, Pandas, NumPy, and Streamlit to analyze academic performance, attendance, grades, and rankings.

Spectrum Technical Services LLC

Technical Intern

May 2025 – Jun 2025

- Supported industrial automation and control-system activities involving SCADA, PI Server, RTUs, and CCTV architectures in oil and gas environments.
- Created 3 technical documents and presentations while collaborating with engineers to analyze automation workflows and operational processes.

Krutanic Solutions

Full-Stack Intern

Nov 2024 – Dec 2024

- Built and enhanced responsive web applications using React, JavaScript, and MongoDB.
- Strengthened UI consistency across 10 components, enhancing user experience while maintaining design standards.
- Collaborated with a remote development team to deliver 3 full-stack features within project timelines.

EDUCATION

Rajagiri School of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering | 7.05 CGPA

2022 – 2026

Indian School Al Wadi Al Kabir

Senior Secondary School (XII-CBSE) | 83%

2020 – 2022

High School (X-CBSE) | 91%

2019 – 2020

PROJECTS

Hybrid RAG System: Engineered an Agentic Hybrid RAG pipeline combining BM25 keyword retrieval and FAISS semantic search, achieving a 25% improvement in overall evaluation score over FAISS-only retrieval across 12 test queries.

Corporate Device Control Suite (CDCS): Built a centralized Linux endpoint management platform with monitoring, policy enforcement, and AI-based ticketing to strengthen endpoint control and reduce manual intervention.

Rajagiri Event Management System (REMS): Designed an event management platform for students and faculty, streamlining event registration and coordination processes.

PUBLICATION

Corporate Device Control Suite: A Centralized Linux Endpoint Governance System – IEEE International Conference on Computing, Communication, Security and Intelligent Systems (IC3SIS 2026), IEEE Xplore Digital Library, 2026.

DOI: 10.1109/IC3SIS69949.2026.11608294

CERTIFICATIONS

IBM SkillsBuild – Cybersecurity Fundamentals

ACHIEVEMENTS

HackS'US Edition V (2026) – 2nd Place

- Contributed as an Audio ML Engineer in developing a system for analyzing and classifying music segments.

HackS'US Edition IV (2025) – Best Feasible Solution

- Implemented a smart commute platform for organizations focused on increasing travel efficiency and sustainability.